

SIGN PATTERNS OF REAL POWERS OF THE GÖLLNITZ–GORDON PRODUCT: A COMPLETE RESOLUTION OF A CONJECTURE OF SCHLOSSER AND ZHOU

JAIDEEP SAI PADHI

ABSTRACT. Let $Q_8(q) = (q, q^7; q^8)_\infty / (q^3, q^5; q^8)_\infty$ and write $Q_8(q)^\delta = \sum_{n \geq 0} c_\delta(n) q^n$ for real δ . Schlosser and Zhou conjectured that for $\beta \leq \delta \leq 4$, where $\beta = 2.664479\overline{11} \dots$ is the point at which $c_\delta(14)$ vanishes, the signs of $c_\delta(n)$ follow a fixed pattern of period 8. We show that this holds for every $\delta \in [8/3, 4]$ and all $n \geq 0$, and that it fails for every $\delta \in [\beta, 8/3)$. We also prove the remaining open part of the conjecture, $-1 < \delta < -0.99$, where the coefficients degenerate as $\delta \rightarrow -1$, and give certified proofs of the parts previously treated by He and Li, so that the conjecture is completely resolved. The failure first occurs near $n \approx 7.6 \times 10^5$, far beyond the range of the numerical evidence. The mechanism is that at the residues $n \equiv 2, 6 \pmod{8}$ the leading circle-method term vanishes identically, so the sign is decided by two secondary terms whose exponential growth rates coincide exactly at $\delta = 8/3$. The proof is self-contained. It combines an exact description of the cusp behaviour of Q_8 , obtained from the Weil representation attached to a lattice of level 64, with an explicit Hardy–Ramanujan–Rademacher expansion and certified computations in exact and ball arithmetic.

1. INTRODUCTION

For real δ let

$$Q_8(q) = \frac{(q, q^7; q^8)_\infty}{(q^3, q^5; q^8)_\infty}, \quad Q_8(q)^\delta = \exp(\delta \log Q_8(q)) = \sum_{n \geq 0} c_\delta(n) q^n,$$

where $\log Q_8(0) = 0$. The function $q^{1/2} Q_8(q)$ is the Göllnitz–Gordon continued fraction. Taking logarithmic derivatives gives the recurrence

$$(1) \quad n c_\delta(n) = \delta \sum_{k=1}^n b_k c_\delta(n-k), \quad b_k = \sum_{d|k} \epsilon(d) d,$$

where $\epsilon(d) = 1$ for $d \equiv 3, 5$, $\epsilon(d) = -1$ for $d \equiv 1, 7 \pmod{8}$, and $\epsilon(d) = 0$ otherwise. In particular each $c_\delta(n)$ is a polynomial in δ of degree at most n with rational coefficients.

Schlosser and Zhou [11] studied sign patterns of real powers of infinite products and posed a number of conjectures in an appendix. For the Göllnitz–Gordon product they conjectured the following.

Conjecture 1.1 ([11, Conj. 21]). *The q -series coefficients of $Q_8(q)^\delta$ exhibit*

(a) *for $\delta = 2$ the length 16 sign pattern $+-+--+-+--+-+--$;*

(b) *for $2.664479110226972 \dots \approx \beta \leq \delta \leq 4$ the sign pattern*

$$(2) \quad \sigma = (+, -, +, +, -, +, -, -) \quad (n \equiv 0, 1, \dots, 7 \pmod{8}),$$

where β is the unique real root in $(2, 3)$ of $x^{12} - 90x^{11} + 1457x^{10} + 30486x^9 - 537081x^8 + 1892346x^7 - 3683653x^6 - 837509646x^5 + 774767020x^4 + 3333687384x^3 - 40887173664x^2 + 94379731200x + 49816166400$;

- (c) for $-1 < \delta \leq \frac{7-\sqrt{73}}{2} \approx -0.77200187$ the sign pattern $\sigma' = (+, +, +, -, -, -, -, +)$;
 (d) for $\delta = -2$ the length 16 sign pattern $+++++-----++++--$.

As in [11], where the analogous Conjecture 2 is phrased as non-negativity of the coefficients of a dissection, we read a sign pattern as allowing zero coefficients. Exactly three zeros occur in the ranges above: $c_4(3) = 0$, $c_{-2}(8) = 0$, and $c_b(4) = 0$ at the endpoint $b = \frac{7-\sqrt{73}}{2}$ (which is where the range in (c) comes from).

Parts (a), (d), and part (c) for $-0.99 \leq \delta \leq \frac{7-\sqrt{73}}{2}$ were proved by He and Li [2], with finite checks done in floating point. We resolve what remained, and give certified proofs of all parts.

Theorem 1.2. *For every $\delta \in [8/3, 4]$ and every $n \geq 0$, $c_\delta(n)$ has the sign $\sigma(n \bmod 8)$. The only vanishing coefficient is $c_4(3) = 0$.*

Theorem 1.3. *Let $2 < \delta < 8/3$. There is $n_1(\delta)$ such that $c_\delta(n) < 0$ for all $n \geq n_1(\delta)$ with $n \equiv 10 \pmod{16}$, and $c_\delta(n) > 0$ for all $n \geq n_1(\delta)$ with $n \equiv 14 \pmod{16}$. For example,*

$$c_{2.66448}(1\,000\,010) < -5.7 \cdot 10^{387}.$$

Theorem 1.4. *Parts (a), (c) and (d) of Conjecture 1.1 hold. In particular, for every $\delta \in (-1, \frac{7-\sqrt{73}}{2}]$ and every $n \geq 0$, $c_\delta(n)$ has the sign $\sigma'(n \bmod 8)$.*

Corollary 1.5. *Conjecture 1.1 is completely resolved: parts (a), (c), (d) are true; part (b) is true for $\delta \in [8/3, 4]$ and false for every $\delta \in [\beta, 8/3)$.*

For $\delta = \beta$ the first failures predicted by the asymptotic expansion are at $n = 759\,390$ (residue $14 \bmod 16$) and $n = 759\,402$ (residue $10 \bmod 16$). The crossover point increases rapidly as $\delta \rightarrow 8/3^-$; for instance it is about 8.3×10^{10} at $\delta = 2.66666$.

The mechanism. The dominant cusps of Q_8 are $3/8$ and $5/8$. Their contributions combine to $2 \cos(3\pi n/4)$ times a Bessel function, which vanishes *identically* for $n \equiv 2, 6 \pmod{8}$. For those residues the sign of $c_\delta(n)$ is decided by two lower-order terms:

- a secondary term at the cusps $3/8, 5/8$, of size $\exp(\frac{\pi}{4}\sqrt{(\delta-2)(2n+\delta)})$, which always has the sign required by (2);
- the leading term at the cusps of denominator 16, of size $\exp(\frac{\pi}{8}\sqrt{\delta(2n+\delta)})$, whose sign in the classes $n \equiv 10, 14 \pmod{16}$ is wrong when $2 < \delta < 3$.

The two rates agree exactly when $\delta - 2 = \delta/4$, that is $\delta = 8/3$. At $\delta = 8/3$ the correct term still wins by a fixed factor $1/0.375$. Because the two rates differ by only about $10^{-3}\sqrt{n}$ near β , the failure is invisible for small n .

Method. The proof is self-contained apart from classical facts: Poisson summation, the generation of $SL_2(\mathbb{Z})$ by S and T , Ford circles, and integral representations of I_1 .

- §2 writes Q_8 as a quotient of theta series.
- §3 determines the behaviour of Q_8 at every cusp. The pair of theta series lies in a finite projective orbit of 48 elements under the Weil representation of $SL_2(\mathbb{Z})$ at level 64; exactly one of its ten cusp types grows, and it has a rigid shape (Lemma 3.4).
- §4 carries out Rademacher's method with fully explicit error terms.
- §5 proves Theorem 1.2, splitting into $n \leq 700$ (exact polynomial arithmetic), $701 \leq n \leq 20000$ (ball arithmetic), and $n > 20000$ (an analytic tail estimate).
- §6 proves Theorem 1.3.
- §8 describes the computations and what they rely on.

Relation to earlier work. For $\delta = 1$ the sign pattern of the coefficients of the Göllnitz–Gordon continued fraction was established by Richmond and Szekeres [10] using Iseki’s transformation formulae [9], and Hirschhorn [8] studied the expansion further. Asymptotics for coefficients of eta-quotients and related products were developed by Chern [7]. Several of the conjectures in [11] have since been proved; see [2] and [6]. He and Li [2] developed a general explicit circle method for real powers of such products and proved several of the conjectures in [11]. For Conjecture 1.1 their method applies at $\delta = 2$ and for some negative δ , but a technical condition fails for $\beta \leq \delta \leq 4$. We do not use results of [2]; the cusp analysis of §3 replaces the general transformation formula used there.

2. THETA-QUOTIENT FORM

Throughout, $e(x) = e^{2\pi i x}$ and $\zeta_m = e(1/m)$. By the Jacobi triple product with base q^8 ,

$$(q^a, q^{8-a}, q^8; q^8)_\infty = \sum_{m \in \mathbb{Z}} (-1)^m q^{4m^2 + (a-4)m}.$$

The factor $(q^8; q^8)_\infty$ cancels in Q_8 . Putting $j = 8m + a - 4$ and $q = e(\tau)$ we obtain

$$(3) \quad Q_8 = e(-\tau/2) \frac{S_1(\tau)}{S_3(\tau)}, \quad S_a(\tau) = \sum_{j \in \mathbb{Z}} \psi_a(j) e(\tau j^2/16),$$

where ψ_a is 16-periodic, with $\psi_a(j) = (-1)^{(j-a+4)/8}$ if $j \equiv a - 4 \pmod{8}$ and $\psi_a(j) = 0$ otherwise.

Notation. For coprime h, k with $k \geq 1$ we write

$$\tau = \frac{h}{k} + \frac{iz}{k^2} \quad (\operatorname{Re} z > 0), \quad M = 16k.$$

Lemma 2.1. *With the principal square root,*

$$S_a\left(\frac{h}{k} + \frac{iz}{k^2}\right) = \frac{1}{16} \sqrt{\frac{8}{z}} \sum_{\ell \in \mathbb{Z}} \mathcal{G}_a(\ell) X^{\ell^2}, \quad X = e^{-\pi/(32z)}, \quad \mathcal{G}_a(\ell) = \sum_{s \bmod M} \psi_a(s) \zeta_M^{hs^2 + \ell s}.$$

Proof. We have $e(\tau j^2/16) = \zeta_M^{hj^2} e^{-\pi z j^2/(8k^2)}$, and $j \mapsto \psi_a(j) \zeta_M^{hj^2}$ is M -periodic. Writing $j = s + Mt$ and applying Poisson summation in t to the Gaussian gives

$$\sum_t e^{-\pi z (s+Mt)^2/(8k^2)} = \frac{1}{M} \sqrt{\frac{8k^2}{z}} \sum_\ell e(\ell s/M) e^{-\pi \ell^2/(32z)}.$$

All series converge absolutely and are holomorphic in z . □

3. CUSP STRUCTURE

For coprime h, k choose $\gamma = \begin{pmatrix} h & b \\ k & d \end{pmatrix} \in SL_2(\mathbb{Z})$. With $\tau = h/k + iz/k^2$ a direct computation gives

$$(4) \quad \tau' = \gamma^{-1}\tau = -\frac{d}{k} + \frac{i}{z}, \quad \operatorname{Im} \tau' = \operatorname{Re}(1/z).$$

3.1. Theta functions of level 64. For $c \in \mathbb{Z}/32\mathbb{Z}$ let $\vartheta_c(\tau) = \sum_{J \equiv c(32)} e(\tau J^2/64)$. Then $\vartheta_c = \vartheta_{-c}$, and comparing with (3),

$$(5) \quad S_1 = \vartheta_6 - \vartheta_{10}, \quad S_3 = \vartheta_2 - \vartheta_{14}.$$

Lemma 3.1. *We have $\vartheta_c(\tau + 1) = e(c^2/64) \vartheta_c(\tau)$ and*

$$\vartheta_c(-1/\tau) = \frac{(i/(32\tau))^{-1/2}}{32} \sum_{d \bmod 32} e(cd/32) \vartheta_d(\tau).$$

Proof. The first identity is clear. For the second, apply Poisson summation to $m \mapsto e^{\pi i \tau (32m+c)^2/32}$. Its Fourier transform at ν is $\frac{1}{32}(-i\tau/32)^{-1/2} e(\nu c/32) e^{-\pi i \nu^2/(32\tau)}$. Grouping ν by residues modulo 32 and replacing τ by $-1/\tau$ gives the claim. $\square$

For a symmetric vector $v = (v_c)_{c \in \mathbb{Z}/32}$ (so $v_c = v_{-c}$) put $v \cdot \vartheta = \sum_c v_c \vartheta_c$. Let

$$D = \text{diag}(\zeta_{128}^{2c^2}), \quad E = (\zeta_{128}^{4cd})_{c,d}.$$

By Lemma 3.1,

$$(v \cdot \vartheta)(T\tau) = (vD) \cdot \vartheta(\tau), \quad (v \cdot \vartheta)(S\tau) = s(\tau) (vE) \cdot \vartheta(\tau),$$

where the scalar $s(\tau)$ does not depend on v . Since $SL_2(\mathbb{Z})$ is generated by S and T , and $S^{-1} = S^3$ while T^{-1} acts projectively as D^{127} (because $D^{128} = I$), every γ is a word $g_1 \cdots g_r$ in S and T without inverses, up to the central element $-I$, which acts on symmetric vectors by a scalar. Iterating,

$$(6) \quad \frac{S_1}{S_3}(\gamma\tau') = R_{(w_1, w_3)}(\tau') := \frac{w_1 \cdot \vartheta(\tau')}{w_3 \cdot \vartheta(\tau')}, \quad (w_1, w_3) = (v_1, v_3) \rho(g_1) \cdots \rho(g_r),$$

where v_1, v_3 are the coefficient vectors in (5) and $\rho(S) = E$, $\rho(T) = D$. The accumulated scalar is the same for both components, so R depends only on the projective class of the pair.

Proposition 3.2. *The projective orbit $\mathcal{O}$ of (v_1, v_3) under right multiplication by D and E has exactly 48 elements and is closed under both maps. Under D it decomposes into 10 cycles, which we call cusp types.*

Proof. This is an exact computation in $\mathbb{Z}[\zeta_{128}] = \mathbb{Z}[x]/(x^{64} + 1)$. Starting from (v_1, v_3) , images under D and E were generated. Each image was tested for projective equality with the known elements by checking all cross-products $p_i q_j = p_j q_i$ exactly. All 96 images coincide with one of 48 elements. $\square$

Near $\text{Im } \tau' = \infty$,

$$w \cdot \vartheta(\tau') = \sum_{J \geq 0} A_J q'^{J^2/64}, \quad q' = e(\tau'), \quad A_0 = w_0, \quad A_J = 2w_{J \bmod 32} \quad (J \geq 1).$$

So the leading exponent of the a -th component is $J_a = \min\{J \geq 0 : w_{a,J} \neq 0\}$. Since w is symmetric, no cancellation between J and $-J$ can occur.

Proposition 3.3. *Exactly one cusp type is growing, i.e. has $J_1 < J_3$, and for it $(J_1, J_3) = (2, 6)$. Of the remaining nine types, one has $(J_1, J_3) = (6, 2)$ and eight have $J_1 = J_3 \in \{0, 1, 2\}$. For the growing type,*

- $\text{supp } w_1 = \{J \equiv \pm 2 \pmod{16}\}$ and $\text{supp } w_3 = \{J \equiv \pm 6 \pmod{16}\}$;
- all nonzero entries of w_1 and w_3 have the same absolute value.

Proof. Exact computation in $\mathbb{Z}[\zeta_{128}]$. The absolute values were compared through the exact products $w_{a,c} \overline{w_{a,c}}$. $\square$

Lemma 3.4 (growing cusps). *Suppose h/k is of growing type and put $t = |q'| = e^{-2\pi \operatorname{Re}(1/z)}$. Then*

$$Q_8(\tau)^\delta = \Phi e^{\pi\delta z/k^2} e^{\pi\delta/z} \left(1 - \delta\beta e^{-2\pi/z} + \mathcal{R}(z)\right), \quad |\Phi| = |\beta| = 1,$$

with $|\mathcal{R}(z)| \leq \tilde{F}_\delta(t) - 1 - \delta t$, where

$$\tilde{F}_\delta = (1 - \bar{U})^{-\delta} (1 - \bar{V})^{-\delta}, \quad \bar{U}(t) = \sum_{\substack{J \equiv \pm 2 \pmod{16} \\ J > 2}} t^{(J^2-4)/64}, \quad \bar{V}(t) = \sum_{\substack{J \equiv \pm 6 \pmod{16} \\ J > 6}} t^{(J^2-36)/64}.$$

Proof. By Proposition 3.3,

$$R = C q'^{-1/2} \frac{1+u}{1+v}, \quad u = \sum_{J>2} \alpha_J q'^{(J^2-4)/64}, \quad v = \sum_{J>6} \beta_J q'^{(J^2-36)/64},$$

with $|C| = |\alpha_J| = |\beta_J| = 1$. The exponents are integers; $u = O(q'^3)$ and $v = \beta_{10} q' + O(q'^7)$. Put $\beta = \beta_{10}$. From (3), (4) and (6),

$$e(-\delta\tau/2) = e(-\delta h/2k) e^{\pi\delta z/k^2}, \quad q'^{-\delta/2} = e(\delta d/2k) e^{\pi\delta/z}, \quad q' = e(-d/k) e^{-2\pi/z}.$$

All unimodular constants are absorbed into Φ and β . For the δ -th power: on the connected set $\{\operatorname{Re} z > 0, \operatorname{Re}(1/z) \geq 1\}$ we have $|u|, |v| < \frac{1}{2}$, so $\delta \log Q_8$ and $\delta(\logarithm \text{ of the displayed product})$, built from principal logarithms of $1+u$ and $1+v$, differ by a continuous function with values in $2\pi i\mathbb{Z}$, hence by a constant; this constant is absorbed into Φ . Expand $(1+u)^\delta (1+v)^{-\delta}$ binomially and use $|\binom{\pm\delta}{r}| \leq \binom{\delta+r-1}{r}$ for $\delta > 0$. Every coefficient is then majorized by the corresponding coefficient of $\tilde{F}_\delta(t)$. The coefficient of t in $\tilde{F}_\delta$ is δ , coming from β_{10} , and all other exponents are integers ≥ 2 . $\square$

Lemma 3.5 (which cusps grow). *If h/k is of growing type, then $4 \mid k$, so for each k at most $\varphi(k) \leq k/2$ fractions h/k are growing. For $k \leq 40$, h/k is growing if and only if $8 \mid k$ and $h \equiv 3, 5 \pmod{8}$.*

Proof. Lemma 2.1 and (6) give the same expansion of S_a in powers of $X = |q'|^{1/64} \cdot (\text{phase})$, so by uniqueness of such expansions $J_a = \min\{\ell \geq 0 : \mathcal{G}_a(\ell) + \mathcal{G}_a(-\ell) \neq 0\}$. By Proposition 3.3, a growing cusp has $J_1 = 2$, so $\mathcal{G}_1(2) \neq 0$ or $\mathcal{G}_1(-2) \neq 0$.

Write $s = r + 8m$ with $r = a - 4$ and $m \bmod 2k$. Then $\psi_a(s) = (-1)^m = e(km/2k)$ and

$$\mathcal{G}_a(\ell) = e\left(\frac{hr^2 + \ell r}{16k}\right) g(8h, 2hr + \ell + k; 2k), \quad g(A, B; K) = \sum_{m \bmod K} e\left(\frac{Am^2 + Bm}{K}\right).$$

We use two elementary facts.

- (i) Let $g_0 = \gcd(A, K)$, $A = g_0 A'$, $K = g_0 K'$. Writing $m = m_0 + K'j$ with $m_0 \bmod K'$ and $j \bmod g_0$, we have $Am^2/K \equiv A'm_0^2/K' \pmod{1}$, so

$$g = \sum_{m_0} e\left(\frac{A'm_0^2}{K'} + \frac{Bm_0}{K}\right) \sum_{j \bmod g_0} e(Bj/g_0).$$

Hence $g = 0$ unless $g_0 \mid B$, and in that case $g = g_0 g(A', B/g_0; K')$.

- (ii) If $\gcd(A, K) = 1$, substituting $m = m' + t$ gives

$$|g|^2 = \sum_t e\left(\frac{At^2 + Bt}{K}\right) \sum_{m'} e\left(\frac{2Atm'}{K}\right) = K(1 + [2 \mid K] e(AK/4 + B/2)).$$

Take $A = 8h$, $K = 2k$, $B = 2hr + \ell + k$. If k is odd then $g_0 = 2$, and $2 \mid B$ forces ℓ to be odd. If $k \equiv 2 \pmod{4}$ then h is odd and $g_0 = 4$; since $2hr \equiv 2$ and $k \equiv 2 \pmod{4}$, the condition $4 \mid B$ forces $\ell \equiv 0 \pmod{4}$. In both cases $\mathcal{G}_1(\pm 2) = 0$, a contradiction. The bound $\varphi(k) \leq k/2$ holds because k is even.

The final statement was verified for all 490 fractions with $k \leq 40$ by exact zero tests of $\mathcal{G}_a(\ell) + \mathcal{G}_a(-\ell)$ modulo the cyclotomic polynomial Φ_{16k} . $\square$

Proposition 3.6 (non-growing cusps). *For every non-growing type and every τ' with $\text{Im } \tau' \geq 1$ we have $|R(\tau')| \leq K_{\text{NG}} := 23.6475$. Consequently, at every non-growing cusp and for every z with $\text{Re}(1/z) \geq 1$,*

$$|Q_8(\tau)^\delta| \leq e^{\pi\delta \text{Re } z/k^2} K_{\text{NG}}^\delta \quad (\delta > 0).$$

Proof. Each R is periodic with period equal to the length $L \in \{2, 4, 8\}$ of its cycle, so it suffices to consider $0 \leq \text{Re } \tau' \leq L$.

- On $\text{Im } \tau' \geq 6$ we use the bound

$$|R| \leq \frac{|A_{1,J_1}|}{|A_{3,J_3}|} t^{(J_1^2 - J_3^2)/64} \frac{1 + \varepsilon_1}{1 - \varepsilon_3}, \quad \varepsilon_a = \sum_{J > J_a} \frac{|A_{a,J}|}{|A_{a,J_a}|} t^{(J^2 - J_a^2)/64},$$

with $t = e^{-2\pi \text{Im } \tau'}$. It is non-increasing in $\text{Im } \tau'$ because $J_1 \geq J_3$, so it suffices to evaluate it at $\text{Im } \tau' = 6$.

- The rectangle $[0, L] \times [1, 6]$ is covered by complex balls on which the theta series, truncated at $J \leq 40$ with an explicit error term $2 \max |A| e^{-2\pi \text{Im } \tau' \cdot 41^2/64}$, are evaluated in ball arithmetic. The rectangles are bisected best-first until every certified upper bound is within a factor 1.02 of the largest sampled value.

The largest certified bound is $23.64743\dots$, while the largest sampled value is 23.184 . The bound for Q_8 follows from (3), (4) and $|e(-\delta\tau/2)| = e^{\pi\delta \text{Re } z/k^2}$. $\square$

3.2. Exact data at the cusps of denominators 8 and 16.

Lemma 3.7. *For every n the following hold.*

- The main-term coefficients $\Phi e(-nh/k)$ of Lemma 3.4 sum to $2 \cos(3\pi n/4)$ over $h/k \in \{3/8, 5/8\}$, and to $4 \cos(\pi n/8) \cos(\pi(n+\delta)/2)$ over $h/k \in \{3/16, 5/16, 11/16, 13/16\}$.
- The correction coefficients $\Phi \beta e(-nh/k)$ sum to $2 \cos((3n-1)\pi/4)$ over $h/k \in \{3/8, 5/8\}$. Consequently the total correction term at $k = 8$ in Proposition 4.4 is $-2\delta \cos((3n-1)\pi/4) \mathcal{P}_8(\delta-2)$.

Proof. Put $c_{h,k} = \widehat{\mathcal{G}}_1(4)/\widehat{\mathcal{G}}_3(36)$, where $\widehat{\mathcal{G}}_a(e) = \sum_{\ell^2=e} \mathcal{G}_a(\ell)$. Exact computation in $\mathbb{Z}[\zeta_{128}]$ and $\mathbb{Z}[\zeta_{256}]$ gives $c_{h,k} = \zeta_{16k}^t$ with

$$t = 24, 40 \quad (h/k = 3/8, 5/8), \quad t = 216, 232, 152, 168 \quad (h/k = 3/16, 5/16, 11/16, 13/16).$$

It also gives $\widehat{\mathcal{G}}_3(100)/\widehat{\mathcal{G}}_3(36) = \rho_h$ with $\rho_3 = e(1/8)$ and $\rho_5 = e(-1/8)$ at $k = 8$. The branch integers $m_{h,k}$ of Lemma 3.8 are $0, 0, 0, 0, 1, 1$ in the order $3/8, 5/8, 3/16, 5/16, 11/16, 13/16$. Hence the δ -dependent phase of the main term,

$$\delta(\text{Arg } c_{h,k} + 2\pi m_{h,k} - \pi h/k),$$

equals 0 at $k = 8$, equals $-\pi\delta/2$ at $h = 3, 5$ with $k = 16$, and equals $+\pi\delta/2$ at $h = 11, 13$ with $k = 16$. Summing complex-conjugate pairs gives (i).

For (ii), by Lemma 2.1 the first correction in $(S_1/S_3)^\delta$ is $-\delta\rho_h X^{64} = -\delta\rho_h e^{-2\pi/z}$, so $\beta_{h,8} = \rho_h$, while $\Phi_{h,8} = 1$ by (i). Summing over $h = 3, 5$ gives $e(-3n/8 + 1/8) + e(-5n/8 - 1/8) = 2 \cos((3n-1)\pi/4)$. $\square$

Lemma 3.8 (branch constants). *Let $\mathcal{D} = \{z : \operatorname{Re} z > 0, \operatorname{Re}(1/z) \geq 1\}$ and let $\log Q_8 = \sum_{m \geq 1} b_m q^m / m$. For each h/k in Lemma 3.7 there is an integer $m_{h,k}$ such that, for all $z \in \mathcal{D}$,*

$$\log Q_8(\tau) = -\pi i \tau + \operatorname{Log} c_{h,k} - 32 \log X + \operatorname{Log}(1+u) - \operatorname{Log}(1+v) + 2\pi i m_{h,k},$$

where $\log X = -\pi/(32z)$. The integers $m_{h,k}$ are those listed in the proof of Lemma 3.7.

Proof. The set $\mathcal{D}$ is connected. On it $t = |q'| \leq e^{-2\pi}$, so by the proof of Lemma 3.4, $|u| \leq \bar{U}(e^{-2\pi}) < 10^{-8}$ and $|v| \leq \bar{V}(e^{-2\pi}) < 2 \cdot 10^{-3}$; hence every principal logarithm on the right is continuous. The difference of the two sides is therefore a continuous function with values in $2\pi i \mathbb{Z}$, hence constant. At $z = 1$ it was evaluated in 160-bit ball arithmetic. The series was summed over $m \leq 8000$, with the tail bounded using $|b_m|/m \leq 1 + \ln m$, and each term q^m was computed directly as $e^{2\pi i m \tau}$. In each case the result equals the stated integer to within $3 \cdot 10^{-11}$. $\square$

4. THE EXPLICIT CIRCLE METHOD

For $k \geq 1$, $B > 0$ and $y = \sqrt{2n + \delta}$ put

$$\mathcal{P}_k(B) = \frac{2\pi}{k} \frac{\sqrt{B}}{y} I_1\left(\frac{2\pi}{k} \sqrt{B} y\right), \quad \text{and} \quad \mathcal{P}_k(B) = 0 \quad \text{for } B \leq 0.$$

Lemma 4.1. *Let K be the circle $|z - \frac{1}{2}| = \frac{1}{2}$, oriented negatively. Then for $B > 0$,*

$$\frac{i}{k^2} \int_K \exp\left(\frac{\pi B}{z} + \frac{\pi(2n + \delta)z}{k^2}\right) dz = \mathcal{P}_k(B).$$

Proof. The map $w = 1/z$ sends K to the line $\operatorname{Re} w = 1$. The claim follows from the integral representation $I_1(x) = \frac{x/2}{2\pi i} \int_{1-i\infty}^{1+i\infty} e^{s+x^2/4s} s^{-2} ds$ with $s = \pi B w$. $\square$

Remark 4.2. As a consistency check of Lemma 4.1, of the orientation conventions, and of Lemma 3.7, the sum of the main terms at $k = 8$ and $k = 16$ reproduces the exact coefficients, e.g. $c_{3,2}(488) = 6.6426668 \cdot 10^{16}$ against $6.6426668 \cdot 10^{16}$, and $c_{3,2}(490) = 4.8976498 \cdot 10^9$ against $4.8976517 \cdot 10^9$; the difference is of the size of the omitted terms.

Lemma 4.3 (Rademacher path). *Let $N \geq 1$ and let $\frac{h_1}{k_1} < \frac{h}{k} < \frac{h_2}{k_2}$ be consecutive in the Farey sequence of order N , taken cyclically on $[0, 1)$. Let $\gamma_{h,k}$ be the upper arc of the Ford circle $C(h/k)$, which has centre $\frac{h}{k} + \frac{i}{2k^2}$ and radius $\frac{1}{2k^2}$, between its points of tangency with $C(h_1/k_1)$ and $C(h_2/k_2)$.*

(a) *The arcs $\gamma_{h,k}$ form a path from some τ_0 to $\tau_0 + 1$, and*

$$c_\delta(n) = \sum_{k \leq N} \sum_h \frac{i}{k^2} e(-nh/k) \int_{z'}^{z''} Q_8\left(\frac{h}{k} + \frac{iz}{k^2}\right)^\delta e^{2\pi n z/k^2} dz,$$

where each integral runs along the arc of K through $z = 1$ from $z' = \frac{k^2 + ikk_1}{k^2 + k_1^2}$ to $z'' = \frac{k^2 - ikk_2}{k^2 + k_2^2}$.

(b) *We have $|z'|, |z''| \leq \sqrt{2}k/(N+1)$ and $\operatorname{Re} z', \operatorname{Re} z'' \leq 2k^2/(N+1)^2$.*

(c) *On the chord $s_{h,k} = [z', z'']$ we have $\operatorname{Re}(1/z) \geq 1$, $0 < \operatorname{Re} z \leq 2k^2/(N+1)^2$, and $|s_{h,k}| \leq 2\sqrt{2}k/(N+1)$.*

(d) *On the two arcs of K from 0 to z' and from z'' to 0 we have $\operatorname{Re}(1/z) = 1$ and $\operatorname{Re} z \leq 2k^2/(N+1)^2$, and each has length at most $\frac{\pi}{2} \cdot \sqrt{2}k/(N+1)$.*

Proof. (a) Adjacent Ford circles are tangent, so consecutive arcs join, and the integrand is holomorphic in $\mathbb{H}$ and 1-periodic. The change of variable $z = -ik^2(\tau - h/k)$ maps $C(h/k)$ onto K , sending its top point $\frac{h}{k} + \frac{i}{k^2}$ to $z = 1$. The point of tangency with $C(h_1/k_1)$ is

$$\frac{h}{k} - \frac{k_1}{k(k^2 + k_1^2)} + \frac{i}{k^2 + k_1^2},$$

which maps to z' ; similarly for z'' . Traversing the upper arc from left to right corresponds to going around K clockwise from z' through 1 to z'' . Finally $d\tau = \frac{i}{k^2} dz$ and $e(-n\tau) = e(-nh/k) e^{2\pi n z/k^2}$.

(b) Farey neighbours satisfy $k + k_i \geq N + 1$, hence $k^2 + k_i^2 \geq (N + 1)^2/2$. Therefore $|z'| = k(k^2 + k_1^2)^{-1/2} \leq \sqrt{2}k/(N + 1)$ and $\operatorname{Re} z' = k^2/(k^2 + k_1^2) \leq 2k^2/(N + 1)^2$, and likewise for z'' .

(c) The chord lies in the closed disc bounded by K , on which $\operatorname{Re}(1/z) \geq 1$. Along a segment, $\operatorname{Re} z$ is at most its value at an endpoint, and $|s_{h,k}| \leq |z'| + |z''|$.

(d) On K we have $\operatorname{Re}(1/z) = 1$. An arc of K from 0 to a point w has length at most $\frac{\pi}{2}|w|$, and $\operatorname{Re} z$ is monotone along it. $\square$

If $N \geq \sqrt{4\pi n + 2\pi\delta}$, then on all the paths in (c) and (d),

$$(7) \quad |e^{(2\pi n + \pi\delta)z/k^2}| \leq e^{(4\pi n + 2\pi\delta)/N^2} \leq e.$$

Proposition 4.4 (explicit expansion). *Let $2 < \delta \leq 4$, $n \geq 1$ and $N \geq \sqrt{4\pi n + 2\pi\delta}$. Then*

$$c_\delta(n) = \sum_{\substack{h/k \text{ growing} \\ k \leq N}} e\left(-\frac{nh}{k}\right) \Phi_{h,k}(\mathcal{P}_k(\delta) - \delta\beta_{h,k}\mathcal{P}_k(\delta - 2)) + E_\delta(n), \quad |\Phi_{h,k}| = |\beta_{h,k}| = 1,$$

and $|E_\delta(n)| \leq \mathcal{E}(\delta, N)$. Here, with $\Sigma_N = (\frac{1}{2} + \frac{N}{8})/(N + 1)$,

$$\mathcal{E}(\delta, N) = \sqrt{2}\pi \Sigma_N e^{1+\pi\delta}(1 + \delta e^{-2\pi}) + 2\sqrt{2}\Sigma_N e^{1+\pi\delta}(\tilde{F}_\delta(e^{-2\pi}) - 1 - \delta e^{-2\pi}) + 2\sqrt{2}e K_{\text{NG}}^\delta.$$

Proof. We start from Lemma 4.3(a).

Growing cusps, main terms. Insert Lemma 3.4. The terms 1 and $-\delta\beta e^{-2\pi/z}$ produce the exponentials $\exp(\pi\delta/z + \pi(2n + \delta)z/k^2)$ and $\exp(\pi(\delta - 2)/z + \pi(2n + \delta)z/k^2)$. Completing each integral from the arc $z' \rightarrow 1 \rightarrow z''$ to all of K and applying Lemma 4.1 gives $\mathcal{P}_k(\delta)$ and $\mathcal{P}_k(\delta - 2)$. By Lemma 4.3(d) and (7), the two added arcs cost at most $\frac{1}{k^2} \cdot 2 \cdot \frac{\pi}{2} \cdot \frac{\sqrt{2}k}{N+1} e^{1+\pi\delta}(1 + \delta e^{-2\pi})$ per growing cusp.

Counting. By Lemma 3.5 there are 2 growing fractions with $k = 8$, 4 with $k = 16$, none with other $k \leq 23$, and at most $k/2$ for each $k \geq 24$ with $4 \mid k$. There are at most $N/4$ such $k \leq N$, so $\sum_{\text{growing}} 1/k \leq \frac{1}{2} + \frac{N}{8}$. This gives the first term of $\mathcal{E}$.

Growing cusps, remainder. Move the $\mathcal{R}$ -integral to the chord, which is allowed by Cauchy's theorem since the integrand is holomorphic in $\operatorname{Re} z > 0$. Put $w = \operatorname{Re}(1/z) \geq 1$. Then

$$|e^{\pi\delta/z}\mathcal{R}(z)| \leq e^{\pi\delta w} \sum_{j \geq 2} \varphi_j e^{-2\pi j w}, \quad \varphi_j \geq 0.$$

Each term is non-increasing in w because $\delta \leq 4$, so the right side is maximal at $w = 1$. The chord length from Lemma 4.3(c) and the same count give the second term.

Non-growing cusps. Move the integral to the chord. There Proposition 3.6 and (7) bound the integrand by $e K_{\text{NG}}^\delta$. Summing $\frac{1}{k^2} \cdot \frac{2\sqrt{2}k}{N+1}$ over at most $\varphi(k) \leq k$ values of h and over $k \leq N$ gives the third term. $\square$

Remark 4.5. The constant $\mathcal{E}(\delta, N)$ is increasing in δ , because $\tilde{F}_\delta = \exp(\delta W)$ where W has non-negative coefficients, and it is decreasing in N . Numerically,

$$\mathcal{E}(8/3, 81) = 4.2 \cdot 10^4, \quad \mathcal{E}(3, 81) = 1.2 \cdot 10^5, \quad \mathcal{E}(4, 81) = 2.9 \cdot 10^6.$$

5. PROOF OF THEOREM 1.2

Fix $\delta \in [8/3, 4]$ and $N = \lceil \sqrt{4\pi n + 2\pi\delta} \rceil$. Using Lemma 3.7 for $k \in \{8, 16\}$ and $|\Phi| = |\beta| = 1$ for $k \geq 24$, Proposition 4.4 gives

$$c_\delta(n) = \mathcal{M}_8 + \mathcal{C}_8 + \mathcal{T},$$

where

$$\mathcal{M}_8 = 2 \cos\left(\frac{3\pi n}{4}\right) \mathcal{P}_8(\delta), \quad \mathcal{C}_8 = -2\delta \cos\left(\frac{(3n-1)\pi}{4}\right) \mathcal{P}_8(\delta-2),$$

and

$$(8) \quad |\mathcal{T}| \leq 4 \underbrace{\left| \cos \frac{\pi n}{8} \cos \frac{\pi(n+\delta)}{2} \right| \mathcal{P}_{16}(\delta) + 4\delta \mathcal{P}_{16}(\delta-2)}_{T_{16}} + \underbrace{\sum_{\substack{24 \leq k \leq N \\ 4|k}} \frac{k}{2} (\mathcal{P}_k(\delta) + \delta \mathcal{P}_k(\delta-2))}_{T_{\geq 24}} + \mathcal{E}(\delta, N).$$

Proposition 5.1 (sign criterion). *Let $\delta \in [8/3, 4]$.*

- (a) *If $n \equiv 0, 1, 3, 4, 5, 7 \pmod{8}$, then $|\mathcal{M}_8| \geq \sqrt{2} \mathcal{P}_8(\delta)$ and $\text{sgn } \mathcal{M}_8 = \sigma(n)$. Hence $\text{sgn } c_\delta(n) = \sigma(n)$ whenever $\sqrt{2} \mathcal{P}_8(\delta) - 2\delta \mathcal{P}_8(\delta-2) > T_{16} + T_{\geq 24} + \mathcal{E}$.*
- (b) *If $n \equiv 2, 6 \pmod{8}$, then $\mathcal{M}_8 = 0$, $|\mathcal{C}_8| = \sqrt{2} \delta \mathcal{P}_8(\delta-2)$ and $\text{sgn } \mathcal{C}_8 = \sigma(n)$. Hence $\text{sgn } c_\delta(n) = \sigma(n)$ whenever $\sqrt{2} \delta \mathcal{P}_8(\delta-2) > T_{16} + T_{\geq 24} + \mathcal{E}$.*

Proof. This follows directly from the explicit values of the cosines: $2 \cos(3\pi n/4) \in \{\pm 2, \pm\sqrt{2}\}$ with sign $\sigma(n)$ on the residues in (a), and on the residues in (b) we have $-2\delta \cos((3n-1)\pi/4) = +\sqrt{2}\delta$ for $n \equiv 2$ and $-\sqrt{2}\delta$ for $n \equiv 6$. $\square$

5.1. The critical comparison. Let $n \equiv 2, 6 \pmod{8}$. Then $|\cos(\pi n/8)| = \frac{\sqrt{2}}{2}$ and $|\cos(\pi(n+\delta)/2)| = |\cos(\pi\delta/2)|$, so

$$(9) \quad \frac{4 \left| \cos \frac{\pi n}{8} \cos \frac{\pi(n+\delta)}{2} \right| \mathcal{P}_{16}(\delta)}{|\mathcal{C}_8|} = \frac{|\cos(\pi\delta/2)|}{\sqrt{\delta(\delta-2)}} \cdot \frac{I_1\left(\frac{\pi}{8}\sqrt{\delta}y\right)}{I_1\left(\frac{\pi}{4}\sqrt{\delta-2}y\right)}.$$

For $\delta \geq 8/3$ we have $\frac{\pi}{8}\sqrt{\delta} \leq \frac{\pi}{4}\sqrt{\delta-2}$, with equality exactly at $\delta = 8/3$. Since I_1 is increasing, the Bessel quotient is then at most 1. Moreover

$$\sup_{[8/3, 4]} \frac{|\cos(\pi\delta/2)|}{\sqrt{\delta(\delta-2)}} = 0.375,$$

attained at $\delta = 8/3$. On $[8/3, 3]$ the function is decreasing, and on $[3, 4]$ it is below 0.375; the latter was certified by bisection into 4000 subintervals.

5.2. An envelope for I_1 .

Lemma 5.2. *The function $x \mapsto \sqrt{2\pi x} e^{-x} I_1(x)$ is increasing on $(0, \infty)$ and tends to 1. Consequently $I_1(x) \leq U(x) := e^x / \sqrt{2\pi x}$ for all $x > 0$, and $I_1(x) \geq \kappa(x_0) U(x)$ for $x \geq x_0$, where $\kappa(x_0) = \sqrt{2\pi x_0} e^{-x_0} I_1(x_0)$.*

Proof. Start from Poisson's integral $I_1(x) = \frac{x}{\pi} \int_{-1}^1 \sqrt{1-t^2} e^{xt} dt$ and substitute $s = x(1-t)$:

$$\sqrt{x} e^{-x} I_1(x) = \frac{1}{\pi} \int_0^{2x} \sqrt{s} \sqrt{2-s/x} e^{-s} ds.$$

The integrand is non-decreasing in x and the interval of integration grows, so the left side is increasing. By monotone convergence its limit is $\frac{\sqrt{2}}{\pi} \Gamma(\frac{3}{2}) = 1/\sqrt{2\pi}$. $\square$

5.3. The range $701 \leq n \leq 20000$. Divide $[8/3, 3]$ and $[3, 4]$ into 16 equal subintervals each. On a subinterval $[a, b]$, every quantity in Proposition 5.1 is bounded by monotonicity, using endpoint values:

- $\mathcal{P}_k(B)$ is bounded below by taking $\sqrt{B}$ minimal, $1/y$ at $y = \sqrt{2n+b}$ and the Bessel argument at $y = \sqrt{2n+a}$, and bounded above symmetrically;
- $\mathcal{E}$ is evaluated at $\delta = b$, by Remark 4.5, with any admissible N ; we take $N = \lceil \sqrt{4\pi n + 2\pi b} \rceil$;
- $|\cos(\pi\delta/2)|$ is monotone on each of $[8/3, 3]$ and $[3, 4]$;
- $T_{\geq 24}$ is bounded by $\#\{24 \leq k \leq N : 4 \mid k\} \cdot \pi(\sqrt{\delta} I_1(\frac{2\pi}{24}\sqrt{\delta}y) + \delta\sqrt{\delta-2} I_1(\frac{2\pi}{24}\sqrt{\delta-2}y))/y$.

The inequalities of Proposition 5.1 were verified in ball arithmetic at 100 bits for every $n \in [701, 20000]$ and every subinterval, with no exceptions.

5.4. The range $n > 20000$. Now $y \geq 200$. Divide every term of (8) by the dominant term: $|\mathcal{C}_8|$ in case (b) and $\sqrt{2}\mathcal{P}_8(\delta)$ in case (a). Apply Lemma 5.2 with $x_0 = \frac{\pi}{8}\sqrt{2/3} \cdot 200$, for which $\kappa(x_0) = 0.99412\dots$. Every quotient other than (9) is then at most $Cy^p e^{-gy}$ with $p \leq \frac{3}{2}$ and $g > 0$, which is decreasing for $y \geq 200$. The constants are evaluated at the worst endpoints of $[8/3, 4]$, using that $\frac{\pi}{4}\sqrt{\delta-2} - \frac{\pi}{8}\sqrt{\delta}$, $\frac{\pi}{4}\sqrt{\delta-2} - \frac{\pi}{12}\sqrt{\delta}$ and $\frac{\pi}{8}\sqrt{\delta}$ are increasing in δ while $\frac{\pi}{4}(\sqrt{\delta} - \sqrt{\delta-2})$ is decreasing. At $y = 200$, in ball arithmetic:

case	$k = 16$ main	$k = 16$ correction	$T_{\geq 24}$	$\mathcal{E}$
(b) $n \equiv 2, 6$	≤ 0.375	$1.4 \cdot 10^{-7}$	$2.2 \cdot 10^{-8}$	$1.3 \cdot 10^{-46}$
(a) other n		$1.5 \cdot 10^{-43}$	$8.2 \cdot 10^{-64}$	$5.0 \cdot 10^{-102}$

In case (a) the quotient $2\delta\mathcal{P}_8(\delta-2)/(\sqrt{2}\mathcal{P}_8(\delta))$ is at most $5.2 \cdot 10^{-40}$. The exponential gaps are at least 0.0859 and 0.118 in case (b) and at least $\frac{\pi}{4}(2 - \sqrt{2})$ in case (a). In both cases the total is less than 1 for all $y \geq 200$.

5.5. The range $n \leq 700$. Each $c_\delta(n) \in \mathbb{Q}[\delta]$ was computed exactly from (1). For each $n \leq 700$:

- the signs at $\delta = 8/3$ and at $\delta = 4$ were evaluated exactly;
- the absence of roots in $(8/3, 4)$ was certified by Descartes' rule of signs applied to $(1+s)^{\deg} c_{8/3+\frac{4/3}{1+s}}(n)$, bisecting whenever the number of sign variations exceeded 1.

For every $n \leq 700$ and every $\delta \in [8/3, 4]$ the sign is $\sigma(n \bmod 8)$, the only zero being $c_4(3) = 0$. The same procedure applied to $[\beta^+, 8/3]$ with $\beta^+ = 2.664479110226973 > \beta$ shows that the signs are $\sigma(n \bmod 8)$ for all $n \leq 700$ and $\delta \in [\beta^+, 8/3]$; applied to $[2.6, 8/3]$ it finds exactly one root, namely that of $c_\delta(14)$ at $\delta = \beta$.

Proof of Theorem 1.2. Combine §5.5 for $n \leq 700$, §5.3 for $701 \leq n \leq 20000$, and §5.4 for $n > 20000$. $\square$

6. PROOF OF THEOREM 1.3

Let $2 < \delta < 8/3$ and $n \equiv 10 \pmod{16}$. Then $\mathcal{M}_8 = 0$ and $\mathcal{C}_8 = +\sqrt{2}\delta\mathcal{P}_8(\delta - 2) > 0$. By Lemma 3.7, the main term at $k = 16$ equals

$$4 \cos \frac{5\pi}{4} \cos \frac{\pi(10 + \delta)}{2} \mathcal{P}_{16}(\delta) = 2\sqrt{2} \cos \left(\frac{\pi\delta}{2} \right) \mathcal{P}_{16}(\delta),$$

which is negative. Since $\delta/4 > \delta - 2$, the quotient (9) tends to $+\infty$ as $n \rightarrow \infty$. Every other term of Proposition 4.4 has Bessel argument at most $\max(\frac{\pi}{4}\sqrt{\delta - 2}, \frac{\pi}{12}\sqrt{\delta})y < \frac{\pi}{8}\sqrt{\delta}y$, or is $O(1)$. Hence $c_\delta(n) < 0$ for large n . The case $n \equiv 14 \pmod{16}$ is identical with all signs reversed.

For the explicit example, take $\delta = 2.66448 \in [\beta, 8/3)$ and $n = 1\,000\,010$. Proposition 4.4, with the exact terms at $k = 8$ and $k = 16$ and absolute bounds for all others, evaluated in 200-bit ball arithmetic, gives

$$c_{2.66448}(1\,000\,010) \leq -5.739 \cdot 10^{387}.$$

This is 13.3% of the $k = 16$ main term. Since $\sigma(1\,000\,010 \bmod 8) = \sigma(2) = +$, this contradicts (2).

Independent confirmation. The direct computation of §8 (`q8_lite.py`, 2600-bit ball arithmetic, no use of §§3–4) evaluates all $c_{2.66448}(n)$, $n \leq 860\,000$, with every sign certified. It finds 12 499 violations of (2), all in the classes $n \equiv 10, 14 \pmod{16}$; the first is $c_{2.66448}(760\,014) > 0$ (with $760\,014 \equiv 14 \pmod{16}$, where σ requires $-$), and from there on every index in these two classes violates the pattern. The location agrees with the crossover predicted by the asymptotic expansion. $\square$

 7. THE REMAINING PARTS: $\delta = \pm 2$ AND $-1 < \delta \leq \frac{7-\sqrt{73}}{2}$

7.1. Negative exponents. For $\delta = -s < 0$ we have $Q_8^{-s} = e(s\tau/2)(S_3/S_1)^s$. The roles of the cusp types are exchanged: the *growing* type is now $(J_1, J_3) = (6, 2)$ of Proposition 3.3, for which, by exact computation, $\text{supp } w_3 = \{J \equiv \pm 2 \pmod{16}\}$, $\text{supp } w_1 = \{J \equiv \pm 6 \pmod{16}\}$, and all nonzero entries have equal modulus. Hence Lemma 3.4 holds verbatim for R^{-1} with s in place of δ . Its correction term $e^{\pi(s-2)/z}$ is bounded for $s \leq 2$ and is left in the remainder. The argument of Lemma 3.5 with $a = 3$ shows that growing cusps have $4 \mid k$, and an exact check for $k \leq 40$ gives exactly $8 \mid k$, $h \equiv 1, 7 \pmod{8}$. The certified bound of Proposition 3.6 for the other types is $\sup |R^{-1}| \leq K'_{\text{NG}} = 23.6475$. The exact data at $k = 8, 16$ (Gauss sums in $\mathbb{Z}[\zeta_{128}]$, $\mathbb{Z}[\zeta_{256}]$ and certified branch integers) give, with $y = \sqrt{2n - s}$, the amplitudes

$$2 \cos \frac{\pi(n - s)}{4} \quad (k = 8), \quad 2 \cos \frac{\pi(n - s)}{8} + 2 \cos \frac{\pi(7n + s)}{8} \quad (k = 16).$$

The analogue of Proposition 4.4 holds for $0 < s \leq 2$ with a single main term per growing cusp and error

$$\mathcal{E}'(s, N) = \sqrt{2}\pi \Sigma_N e^{1+\pi s} + 2\sqrt{2} \Sigma_N e^{1+\pi s} (\tilde{F}_s(e^{-2\pi}) - 1) + 2\sqrt{2} e K'_{\text{NG}}{}^s, \quad N \geq \sqrt{4\pi n} + 1,$$

since $e^{\pi s w} (\tilde{F}_s(e^{-2\pi w}) - 1)$ is non-increasing in w for $s \leq 2$. For part (a), $\delta = 2$, the same modification (no correction term extracted) is used in Proposition 4.4.

7.2. The degeneration at $\delta = -1$. By a theorem of Andrews and Bressoud [5] (see also Richmond–Szekeres [10, Thm. 5.1]), the coefficients of $Q_8(q)^{-1} = (q^3, q^5; q^8)_\infty / (q, q^7; q^8)_\infty$ vanish at all $n \equiv 3 \pmod{4}$. Consequently, at these n the leading amplitude $2 \cos(\pi(n - s)/4) = \mp 2 \sin(\pi\varepsilon/4)$, $\varepsilon = 1 - s = \delta + 1$, tends to 0, while the error term does not, and the method of §5

alone gives a threshold $n_0(\delta) \rightarrow \infty$ as $\delta \rightarrow -1$ (this is the obstruction met in [2]). We remove it by subtracting the expansion at $s = 1$:

$$c_\delta(n) = c_\delta(n) - c_{-1}(n) = \mp 2 \sin \frac{\pi \varepsilon}{4} \mathcal{P}_8(s) + \sum_{k \geq 16} (M_k(s) - M_k(1)) + (E'(s) - E'(1)),$$

and bounding every difference by ε times an explicit quantity, using three lemmas.

Lemma 7.1. *For $|q| < 1$, $|\operatorname{Im} \log Q_8(q)| \leq \frac{\pi}{2} \frac{1}{1-|q|}$. On every Farey chord $|\operatorname{Im} \log Q_8| \leq 0.556 N^2$, and the phase $\theta_{h,k}$ of the main term at a growing cusp (so that $\Phi_{h,k}(s) = e^{is\theta_{h,k}}$) satisfies $|\theta_{h,k}| \leq k^2/2 + 0.01$.*

Proof. $\log Q_8 = \sum_m \epsilon_m \log(1-q^m)$ with principal logarithms and $|\operatorname{Arg}(1-x)| \leq \frac{\pi}{2}|x|$ for $|x| < 1$; sum over odd m . On a chord $\operatorname{Re} z \geq k^2/(2N^2)$, so $1-|q| \geq 0.9\pi/N^2$. For $\theta_{h,k}$ evaluate the branch identity of Lemma 3.8 (for R^{-1}) at $z = 1$, where $1-|q| \geq 0.9 \cdot 2\pi/k^2$. $\square$

Lemma 7.2 (arc lemma). *Let $A_s = \int f_s dz$ over one of the arcs of K from 0 to z' or z'' , with $f_s = \Phi(s) \exp(\pi s/z + (2\pi n - \pi s)z/k^2)$, $s \in [s_0, 1]$, $s_0 > 0$. Then*

$$|\partial_s A_s| \leq e^{1+\pi s} \left[\left(\frac{k^2}{2} + 0.01 \right) \frac{\pi}{\sqrt{2}} \frac{k}{N+1} + \frac{2 + \pi/2}{s_0} + \frac{\pi^2/\sqrt{2}}{k(N+1)} \right].$$

Proof. $\partial_s f_s = (i\theta + \pi w - \pi/(k^2 w))f_s$ with $w = 1/z = 1 + iv$, $|v| \geq k_1/k$. The terms $i\theta f_s$ and $\pi f_s/(k^2 w)$ are bounded using $|f_s| \leq e^{1+\pi s}$ and the arc length. For $\int \pi w f_s dz = -\pi \Phi i e^{\pi s} \int e^{i\pi s v} g(v) dv$, $g = e^{b/(1+iv)}/(1+iv)$, integrate by parts: $|g| \leq e$, and $\int |g'| \leq e \int (b(1+v^2)^{-3/2} + (1+v^2)^{-1}) dv \leq e(1 + \pi/2)$ because $b/(1+v_1^2) = b \operatorname{Re} z' \leq 1$. $\square$

Lemma 7.3. *For $0 < s_0 \leq s \leq 1$ and $y = \sqrt{2n-s}$, $\mathcal{P}_k(s)$ is increasing in s and $\mathcal{P}_k(1) - \mathcal{P}_k(s) \leq (1-s)D_k \mathcal{P}_k(1)$, where $D_k = \frac{1}{2s_0} + \frac{1}{2y^2} + \frac{a_k y}{2\sqrt{s_0}} \left(1 + \frac{1}{a_k \sqrt{s_0} y}\right)$, $a_k = 2\pi/k$.*

Proof. $I_0 = I_2 + 2I_1/x \leq I_1 + 2I_1/x$ gives $0 < I'_1/I_1 = I_0/I_1 - 1/x \leq 1 + 1/x$; differentiate $\log \mathcal{P}_k$. $\square$

Lemma 7.4 (derivatives of the remaining error terms). *Let $s \in [s_0, 1]$ and $N \geq \sqrt{4\pi n} + 1$.*

(i) Growing cusps, chord remainder. *Write $Q_8^{-s} = \Phi(s)e^{\pi s/z - \pi s z/k^2} W^s$ with $W = (1+u)/(1+v)$ as in Lemma 3.4 (for R^{-1}). Then on the chord $s_{h,k}$,*

$$|\partial_s [\Phi(s)e^{\pi s/z - \pi s z/k^2} (W^s - 1)]| \leq e^{\pi s} \left[\left(\frac{k^2}{2} + \frac{2\pi N^2}{k^2} + 0.1 \right) (\tilde{F}_1(e^{-2\pi}) - 1) + \tilde{F}_1(e^{-2\pi}) \ell(e^{-2\pi}) \right],$$

where $\ell = -\log(1 - \bar{U}) - \log(1 - \bar{V})$.

(ii) Non-growing cusps. *On the chord, $|\partial_s Q_8^{-s}| \leq K'_{\text{NG}} (\log K'_{\text{NG}} + 0.01 + 0.556 N^2)$.*

Proof. (i) The derivative equals $(i\theta + \pi/z - \pi z/k^2)\Phi e^{\dots} (W^s - 1) + \Phi e^{\dots} W^s \log W$. On the chord, $|\theta| \leq k^2/2 + 0.01$ by Lemma 7.1; $|1/z| \leq 1/\operatorname{Re} z \leq 2N^2/k^2$ since $\operatorname{Re} z \geq k^2/(2N^2)$; and $\pi|z|/k^2 \leq 0.1$. By the coefficientwise majorization of Lemma 3.4, $|W^s - 1| \leq \tilde{F}_s(t) - 1 \leq \tilde{F}_1(t) - 1$, $|W|^s \leq \tilde{F}_1(t)$ and $|\log W| \leq \ell(t)$, with $t = e^{-2\pi w}$, $w = \operatorname{Re}(1/z) \geq 1$. Both $e^{\pi s w}(\tilde{F}_1(t) - 1)$ and $e^{\pi s w} \tilde{F}_1(t) \ell(t)$ are sums of terms $e^{(\pi s - 2\pi j)w}$ with $j \geq 1$ and non-negative coefficients, hence non-increasing in w since $s \leq 1$, so their maxima are at $w = 1$.

(ii) $\partial_s Q_8^{-s} = -\log Q_8 \cdot Q_8^{-s}$. On the chord $|Q_8^{-s}| \leq K'_{\text{NG}} \leq K'_{\text{NG}}$, while $|\operatorname{Re} \log Q_8| = |\log |Q_8|| \leq \log K'_{\text{NG}} + \pi \operatorname{Re} z/k^2$, using both certified bounds $|R| \leq K_{\text{NG}}$ and $|R^{-1}| \leq K'_{\text{NG}}$ (which are equal to 23.6475). Finally $|\operatorname{Im} \log Q_8| \leq 0.556 N^2$ by Lemma 7.1. $\square$

Summing Lemma 7.4 over the chords, Lemma 7.2 over the arcs (with the counts of Proposition 4.4), Lemma 7.3 and $|\Phi_{h,k}(s) - \Phi_{h,k}(1)| \leq \varepsilon|\theta_{h,k}|$ for the main terms with $k \geq 24$, and $|A_{16}(n, s)| \leq \frac{\pi}{2}\varepsilon$ for the $k = 16$ amplitude, which vanishes at $s = 1$ and has s -derivative at most $\pi/2$, we obtain $|c_\delta(n) \mp 2\sin(\pi\varepsilon/4)\mathcal{P}_8(s)| \leq \varepsilon \cdot B(n)$ with $B(n)$ explicit and independent of ε .

Dividing by ε yields a sufficient condition for the sign at $n \equiv 3, 7 \pmod{8}$ that *does not involve* ε :

$$\frac{\pi}{2} \left(1 - \frac{(\pi\varepsilon_{\max}/4)^2}{6} \right) \mathcal{P}_8(s_0) > \frac{\pi}{2} \mathcal{P}_{16}(1) + \sum_{\substack{24 \leq k \leq N \\ 4|k}} \frac{k}{2} \left(\frac{k^2}{2} + 0.01 + D_k \right) \mathcal{P}_k(1) + (\text{arc}) + (\text{chord}) + (\text{non-growing}).$$

At the other residues $|2\cos(\pi(n-s)/4)| \geq 2 \cdot 0.5698$ for $s \in [0.772, 1]$ and the standard criterion applies.

7.3. Certified verification.

part	$n \leq n_1$ (exact)	$n_1 < n \leq 20000$ (ball arithmetic)	$n > 20000$
(a) $\delta = 2$	integers, $n \leq 400$	criterion holds for $n \geq 295$	ratio $\leq 8.5 \cdot 10^{-13}$
(d) $\delta = -2$	integers, $n \leq 400$	criterion holds for $n \geq 297$	ratio $\leq 8.5 \cdot 10^{-13}$
(c) $s \in [0.772, 1]$	$\mathbb{Q}[\delta]$ on $[-1, -0.772]$, $n \leq 400$	criterion holds for $n \geq 372$	ratio $\leq 4.3 \cdot 10^{-26}$

For (a) and (d), the dominant term at each residue class modulo 16 is the $k = 8$ term, or the $k = 16$ term where the $k = 8$ amplitude vanishes ($n \equiv 2, 6$ for $\delta = 2$; $n \equiv 0, 4 \pmod{8}$ for $\delta = -2$), with the signs of the stated patterns; the exact coefficients for $n \leq 400$ follow the patterns, with the single zero $c_{-2}(8) = 0$. For (c), the finite check divides $c_\delta(n)$ by $\delta + 1$ at the 100 indices $n \equiv 3 \pmod{4}$ where $c_{-1}(n) = 0$, and $c_\delta(4)$ additionally by $\delta^2 - 7\delta - 6 = (\delta - b)(\delta - b')$, which is non-negative on $[-1, b]$; Descartes' rule with bisection then certifies the signs on the closed interval. This proves Theorem 1.4. $\square$

8. COMPUTATIONS

What the computations rely on. Every computational step belongs to one of three categories.

- (1) *Exact integer and rational arithmetic*, in Python integers and FLINT [1] polynomials: Propositions 3.2 and 3.3, Lemma 3.5, the Gauss sums in Lemma 3.7, and §5.5.
- (2) *Ball arithmetic*, in Arb [3] as provided by python-flint 0.9.0 with FLINT 3.6.0 and Python 3.12.3: Proposition 3.6, Lemma 3.8, §§5.3–5.4, and §6. Arb guarantees that each computed ball contains the exact value, and an inequality is accepted only when the whole ball satisfies it.
- (3) *Floating point*, used only for exploration and as hash keys in the orbit computation. Every match found through a hash key was confirmed exactly, and no conclusion depends on floating-point arithmetic.

Scripts.

statement	script
Lemma 3.1 (numerical sanity check)	<code>weil_check.py</code>
Propositions 3.2, 3.3	<code>orbit.py</code> , <code>cuspid_types.py</code>
Lemma 3.5	<code>cuspid_exact.py</code>
Proposition 3.6	<code>supK.py</code>
Lemma 3.7	<code>exact.py</code>
Lemma 3.8	<code>branch_cert.py</code>
Remark 4.5	<code>consts.py</code>
§5.3	<code>certify_mid.py</code>
§5.4	<code>tail_sc.py</code>
§5.5	<code>c3.py</code>
§6	<code>discert_sc.py</code>
Remark ($[\beta^+, 8/3]$, $n \leq 700$)	<code>c3_betaplus.py</code>
§7: negative-exponent cusp data	<code>inv_exact.py</code> , <code>cuspid_exact_inv.py</code> , <code>supKinv.py</code>
§7: part (c)	<code>neg_certify.py</code> , <code>neg_tail772.py</code> , <code>neg_finite2.py</code>
§7: parts (a), (d)	<code>pm2_certify.py</code> , <code>pm2_tail.py</code> , <code>pm2_finite.py</code>
independent direct check of Theorem 1.3	<code>q8_lite.py</code> (or <code>q8_signcheck.py</code>)

The script `q8_lite.py` computes the coefficients of Q_8^δ directly from (1), by a divide-and-conquer evaluation with blocked ball-arithmetic convolutions, and does not use §§3–4. Its ball radii grow like $e^{2.5\sqrt{n}}$, so 2600 bits suffice at $n \approx 8.6 \times 10^5$, with a memory footprint below 1 GB. The older `q8_signcheck.py` uses the Newton exponential, whose radii grow like $e^{6.4\sqrt{n}}$, and needs 15–20 GB.

REFERENCES

- [1] The FLINT team, *FLINT: Fast Library for Number Theory*, version 3.6.0, <https://flintlib.org>.
- [2] B. He and L. Li, *Some conjectures of Schlosser and Zhou on sign patterns of the coefficients of infinite products*, arXiv:2509.10023.
- [3] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), 1281–1292.
- [4] H. Rademacher, *Topics in Analytic Number Theory*, Grundlehren Math. Wiss. 169, Springer, 1973.
- [5] G. E. Andrews and D. M. Bressoud, *Vanishing coefficients in infinite product expansions*, J. Austral. Math. Soc. Ser. A **27** (1979), 199–202.
- [6] K. Bringmann, B. Heim and B. Kane, *On a sign-change conjecture of Schlosser and Zhou*, J. Math. Anal. Appl. **548** (2025).
- [7] S. Chern, *Asymptotics for the Fourier coefficients of eta-quotients*, J. Number Theory **199** (2019), 168–191.
- [8] M. Hirschhorn, *On the expansion of a continued fraction of Gordon*, Ramanujan J. **5** (2001), 369–375.
- [9] S. Iseki, *A partition function with some congruence condition*, Amer. J. Math. **81** (1959), 939–961.
- [10] B. Richmond and G. Szekeres, *The Taylor coefficients of certain infinite products*, Acta Sci. Math. (Szeged) **40** (1978), 347–369.
- [11] M. J. Schlosser and N. H. Zhou, *On the infinite Borwein product raised to a positive real power*, Ramanujan J. **61** (2023), 515–543, doi:10.1007/s11139-021-00519-3; arXiv:2011.10552.

DEPARTMENT OF COMPUTER SCIENCE, PURDUE UNIVERSITY, WEST LAFAYETTE, IN, USA
 Email address: jpadhi@purdue.edu